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Benchmarking Zero-Shot and Prompt-Engineered LLMs for Knowledge Graph Quality Evaluation: A Comparative Study of Llama-3 and GPT-5.4 on FB15k-237 and WN18RR

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Abstract—This study investigates the efficacy of Large Language Models (LLMs) in automating the validation of Knowledge Graphs (KGs). Utilizing two benchmark datasets—FB15k-237 (real-world facts) and WN18RR (lexical relations)—we evaluate Llama-3-70B and GPT-5.4 mini across four distinct prompting strategies: Zero-Shot, Few-Shot, Chain-of-Thought (CoT), and Expert Persona. Our results indicate that basic Zero-Shot prompts exhibit a significant “affirmation bias,” whereas the Expert Persona (S_4) strategy achieves the highest balanced accuracy (71.1% for Llama-3). We further introduce a failure taxonomy to categorize LLM-detected inconsistencies and employ topological analysis to measure the structural impact of LLM-driven graph refinement. Our findings suggest that while LLMs excel at detecting entity-level corruptions, relation-swapping remains a challenge in lexical KGs, highlighting the potential for targeted Supervised Fine-Tuning in specialized KG maintenance.

I. INTRODUCTION

A. Background and Motivation

Knowledge Graphs (KGs)—including prominent open cross-domain examples like Freebase [2], Wikidata [3], DBpedia [1], and YAGO [4]—have become foundational infrastructures for downstream AI applications. Although modern AI often uses KGs for paradigms such as Retrieval-Augmented Generation (RAG), they are widely deployed in enterprise industries [6] to power semantic search [5], [7], personalized recommendation systems [8], [9], [10], conversational agents [11], question answering [20] and business analytics [10].

By applying a graph-based abstraction to data, KGs represent real-world entities of interest as nodes and the relationships between them as directed edges [12]. This structure, frequently expressed as directed edge-labeled graphs (often conceptualized as structured triplets consisting of a subject, predicate, and object, or head, relation, and tail (h, r, t)), provides a concise, intuitive abstraction [12]. Unlike unstructured text corpora—which require complex Natural Language Processing (NLP) [13], [14], [18], [19] and Information Extraction (IE) pipelines [15], [16], [17] to interpret—this explicit graph structure offers a machine-readable foundation capable of supporting deductive logical reasoning and inductive machine learning techniques [12].

However, the utility and “fitness for purpose” of a KG are heavily dependent on its overall quality, which encompasses dimensions such as semantic accuracy, coherency, and coverage [12], [22]. Because KGs are assembled from highly

diverse and dynamic data collections, even state-of-the-art graphs suffer from noise, missing links, and factual corruptions [12], [21]. Initial KGs are inherently incomplete and frequently contain duplicate, contradictory, or incorrect statements [23], [24], [25]. These issues are introduced through various creation methods: collaborative human curation is vulnerable to human error [26], disagreement [27], bias [28], and vandalism [29], while automated extraction can yield imprecise results and entity alignment errors [12]. Consequently, KGs require continuous refinement to maintain their veracity over time [12], [21].

B. The Problem: Limitations of Traditional Validation

Traditionally, the refinement and correction of Knowledge Graphs (KGs) have relied on two distinct pillars: manual human review and automated verification models. While manual human curation is effective for assessing semantic accuracy, it is prohibitively expensive and unscalable for modern KGs containing millions or billions of edges [16], [12].

To automate validation and link prediction, KGs frequently employ Knowledge Graph Embedding (KGE) models, such as the translational models *TransE* [30] (which treats relations as geometric transformations between subject and object vectors) and *RotatE* [31] (which uses relational rotation in complex space). These methods compress the graph into dense, continuous, low-dimensional vector spaces and learn a scoring function to assess the “plausibility” of a given edge based purely on latent structural features.

However, because pure embedding-based methods rely entirely on graph topology and numerical vectors, they lack an interpretable model and struggle to generalize complex semantics [12]. Consequently, they may confidently predict that a link is “likely” based on topological proximity, completely failing to understand the underlying semantic impossibility of the fact itself (e.g., predicting an entity has two distinct birthplaces) [32].

The emergence of Large Language Models (LLMs) represents a paradigm shift in Knowledge Graph (KG) refinement, moving the field away from manual, rule-based, and statistical pipelines toward automated, language-driven, and generative frameworks [33]. LLMs can overcome the traditional bottlenecks by leveraging their advanced semantic understanding and contextual reasoning to autonomously identify, filter, and correct errors [33], [34], [35].

C. Proposed Approach and Contributions

In this study, we move beyond simple zero-shot queries to evaluate the limits of LLM-based KG validation. We perform a head-to-head comparison between a flagship open-source model, Llama-3-70B, and a frontier reasoning model, GPT-5.4 mini. Our methodology tests the hypothesis that sophisticated prompting strategies—specifically Chain-of-Thought (CoT) and Expert Persona—can mitigate the inherent “affirmation bias” often found in LLMs.

The primary contributions of this work are threefold:

- 1) **Multi-Model Benchmarking:** A comparative evaluation of Llama-3 and GPT-5.4 mini performance across four prompting strategies (S_1 to S_4) on two divergent datasets: FB15k-237 (general world facts) and WN18RR (lexical hierarchies).
- 2) **Failure Taxonomy:** The development of a qualitative framework to categorize the specific modes in which LLMs fail to detect KG corruptions, distinguishing between semantic mismatches and ontological violations.
- 3) **Topological Impact Analysis:** A quantitative assessment using graph theory to measure how LLM-driven “cleaning” alters the connectivity and structural integrity of the resulting knowledge base.

II. RELATED WORK

The integration of Large Language Models (LLMs) has introduced several transformative methodologies for the continuous refinement of Knowledge Graphs (KGs).

A. Detecting and Correcting Noise

Automated KG construction often introduces noise, such as misaligned entities and incorrect relations, which degrades the reliability of downstream tasks [35]. LLMs can evaluate the coherence of entity-relationship pairs based on their extensive encoded factual knowledge [35]. Frameworks like *LLM_sim* use LLMs not only to detect erroneous triples but to refine them by generating candidate replacements [35]. The model typically fixes one entity and the relation while predicting a corrected counterpart, ultimately matching the best candidate against the existing KG using contextual similarity.

B. Validating Triple Insertions and Class Memberships

LLMs can act as automated “KG curators” that systematically evaluate new Resource Description Framework (RDF) triples before they are inserted into an existing graph [34]. They address four critical refinement challenges [34]:

- **Violations of predefined classes and properties:** Ensuring new data conforms to the established ontology schema.
- **URI Standardization:** Preventing redundant or duplicated URIs representing the same real-world entity.
- **Semantic Consistency:** Ensuring new triples do not logically contradict existing knowledge within the graph.
- **Syntactic Correctness:** Verifying that triples adhere to strict formatting rules (e.g., possessing exactly one subject, predicate, and object).

Furthermore, by using zero-shot Chain-of-Thought (CoT) prompting with natural language definitions, LLMs can accurately evaluate class membership relations [36]. In evaluations, LLMs have successfully identified actual errors residing in ground-truth KGs, detecting both missing relations and incorrectly asserted ones [36].

C. Knowledge Fusion and Conflict Resolution

At the instance level, KG refinement requires merging heterogeneous knowledge, resolving conflicts, and deduplicating entities [33]. LLMs have evolved from simple lexical matchers into adaptive reasoning agents capable of integrating contextual and structural signals to perform self-correcting fusion [33]. For example, the *Graphfusion* framework utilizes LLMs to merge local sub-graphs by consolidating semantically similar entities and resolving conflicting relationships through source text analysis [37].

D. Standardizing Relational Expressions

The diversity of natural language often results in unaligned extraction, which historically required extensive manual annotation to harmonize [38]. By embedding domain ontologies directly into CoT prompts, LLMs can be guided to map diverse textual expressions into a strict, standardized set of predefined relations [38]. This minimizes relational diversity upfront, reducing the alignment workload during the refinement phase.

E. Bidirectional Collaborative Refinement

To combat LLM limitations like hallucination and long-tail relation bias, recent frameworks pair LLMs with structural Knowledge Graph Embeddings (KGEs) [39]. In the *LEC-KG* framework, this creates a bidirectional refinement loop:

- **KGE to LLM (Structural Feedback):** The KGE evaluates the structural plausibility of candidate triples. If a triple scores poorly, the KGE sends diagnostic signals and textual evidence back to the LLM for re-evaluation [39].
- **LLM to KGE (Semantic Enhancement):** Validated high-confidence triples are fed back to update and improve the KGE’s structural representations [39].

F. Challenges and Open Issues

Despite their potential, employing LLMs for KG refinement faces significant limitations:

- **Semantic Drift:** In long chains of interconnected triples, LLMs can lose coherence and struggle to retain context across multiple steps [34].
- **Complexity and Syntax:** LLMs can struggle with specialized ontological hierarchies and strict serialization formats like Turtle [34].
- **Cost and Scalability:** Deploying large-scale models incurs significant computational costs, limiting real-time scalability [34].

Consequently, current state-of-the-art approaches maintain a “human-in-the-loop” methodology, using LLMs to reduce manual workload while relying on experts to verify complex semantic logic [34].

III. METHODOLOGY

This section details the systematic approach used to evaluate the factual validation capabilities of Llama-3-70B and GPT-5.4 mini. The experiment is framed as a binary classification task where the models must distinguish between valid Knowledge Graph (KG) triplets and those synthetically corrupted to test semantic and factual boundaries.

A. Dataset Selection

We utilize two of the most challenging benchmark KG datasets representing distinct domains of knowledge to ensure the generalizability of our findings:

- **FB15k-237:** A subset of Freebase containing real-world facts. This dataset is a standard benchmark in the KG community, specifically curated to remove redundant inverse relations.
- **WN18RR:** A subset of WordNet representing lexical and hierarchical relationships between words (e.g., hypernyms, member meronyms). This dataset tests linguistic reasoning rather than encyclopedic memory.

B. Data Pre-processing and Sampling

To prepare the datasets for evaluation, we performed a stratified sampling of 1,000 triplets from each source dataset. This sampling ensures a balanced representation of unique relations within the graph. Each selected triplet was cross-referenced against the global dataset to ensure no overlaps between training and evaluation sets. Detailed sampling distributions and counts per relation for FB15k-237 and WN18RR are provided in **Appendix A** and **Appendix B**, respectively.

C. Initial Network Analysis

Before the corruption phase, we conducted a topological analysis of the sampled sub-graphs to characterize their structural complexity. Metrics including node density, average degree, and number of connected components were calculated to establish a baseline for the Knowledge Graph’s connectivity. A summary of these network topological metrics is provided in **Appendix C**.

D. Relation-Aware Synthetic Corruption

A core challenge in KG validation is ensuring that negative samples (lies) are not “obvious” due to simple category errors. We developed a Relation-Aware Corruption Engine to generate a balanced evaluation set ($N = 1000$ per dataset) where each positive triplet is paired with a negative triplet that respects the semantic constraints of the original relation.

1) *Type-Safe Candidate Pools:* Instead of sampling replacements from the global entity list, we implemented a *Per-Relation Profile* strategy. For every relation r in the ontology, we construct two specific candidate sets:

- **Head Pool (S_r^h):** All unique entities appearing in the head position for relation r across the entire dataset.
- **Tail Pool (S_r^t):** All unique entities appearing in the tail position for relation r across the entire dataset.

2) *Modes of Corruption:* For every ground-truth triplet (h, r, t) , the engine generates a negative counterpart using one of three modes:

- **Head Corruption:** Create (h', r, t) where $h' \in S_r^h$ and $h' \neq h$.
- **Tail Corruption:** Create (h, r, t') where $t' \in S_r^t$ and $t' \neq t$.
- **Relation Corruption:** Create (h, r', t) where r' is a random relation sampled from the global ontology pool.

3) *The “Filter Rule” Verification:* To guarantee the integrity of the negative samples, every candidate triplet undergoes a verification check against a Global Truth Set. If a randomly generated corruption inadvertently creates a valid fact, the sample is discarded and regenerated.

4) *Mathematical Representation:* We denote the our Ground Truth Set as $\mathcal{T}$. Each evaluation sample is defined as:

$$\text{Sample} = \begin{cases} (h, r, t, 1, \text{'original'}) & \text{if } (h, r, t) \in \mathcal{T} \\ (h', r, t, 0, \text{'head_corr'}) & \text{if } (h', r, t) \notin \mathcal{T} \\ (h, r, t', 0, \text{'tail_corr'}) & \text{if } (h, r, t') \notin \mathcal{T} \\ (h, r', t, 0, \text{'rel_corr'}) & \text{if } (h, r', t) \notin \mathcal{T} \end{cases} \quad (1)$$

E. Natural Language Mapping

To facilitate LLM inference, raw Knowledge Graph identifiers were mapped to their corresponding natural language labels. This step ensures that the models utilize their internal knowledge rather than attempting to parse machine identifiers. Table I illustrates the transformation of triplets before and after this mapping.

TABLE I
EXAMPLES OF NATURAL LANGUAGE MAPPING FOR FB15K-237 AND WN18RR

Dataset	Raw Triplet (h, r, t)	Mapped Triplet (h, r, t)
FB15k-237	(m.01yjl, /film/actor/film, m.02q_x)	(Tom Hanks, acts in, Toy Story)
WN18RR	(08103417, _hypernym, 08098314)	(Apple, is a type of, Fruit)

F. Prompt Engineering Strategies

The evaluation encompasses four distinct strategies (S_1 – S_4) designed to measure how performance scales with complexity. Actual prompt templates used for each strategy are documented in **Appendix D**.

- **S_1 : Zero-Shot (Baseline):** A direct query asking if the triplet is true or false.
- **S_2 : Few-Shot Learning:** Providing four illustrative examples (two true, two false) before the target query.
- **S_3 : Chain-of-Thought (CoT):** Instructing the model to explain its reasoning step-by-step before a final verdict.
- **S_4 : Expert Persona:** Framing the model as a “Senior Knowledge Graph Engineer” to focus attention on structural integrity.

G. Computational Framework and Parameters

- **Llama-3-70B:** Deployed on the LARCC HPC cluster using a multi-GPU configuration, loaded in 4-bit precision.
- **GPT-5.4 mini:** Accessed via the OpenAI API locally on a macOS environment.
- **Hyperparameters:** Temperature was set to 0 to ensure reproducibility. A maximum completion token limit was set between 150 and 200 tokens to accommodate reasoning.

IV. EXPERIMENTAL RESULTS

This section presents a quantitative evaluation of the two models across all four prompting strategies. We analyze overall performance metrics and break down the models’ ability to detect specific types of Knowledge Graph (KG) corruptions.

A. Overall Performance and Strategy Comparison

Table II summarizes the core metrics—Accuracy, Precision, Recall, and F1-score—for each configuration.

TABLE II
MODEL PERFORMANCE ACROSS PROMPTING STRATEGIES

Dataset	Model/Strategy	Acc.	Prec.	Rec.	F1
FB15k-237	Llama-3 (S_4)	71.1%	67.2%	82.4%	0.740
	Llama-3 (S_3)	61.1%	58.0%	79.8%	0.672
	GPT-5.4 (S_3)	67.8%	66.3%	72.2%	0.691
	GPT-5.4 (S_4)	69.9%	78.3%	55.0%	0.646
WN18RR	Llama-3 (S_4)	69.3%	63.0%	93.9%	0.754
	Llama-3 (S_1)	53.1%	51.6%	97.2%	0.674
	GPT-5.4 (S_4)	70.5%	70.2%	71.3%	0.707
	GPT-5.4 (S_1)	69.6%	71.8%	64.5%	0.680

The overall performance trends across four strategies are visualized in Figure 1, which highlights the scaling behavior of each model.

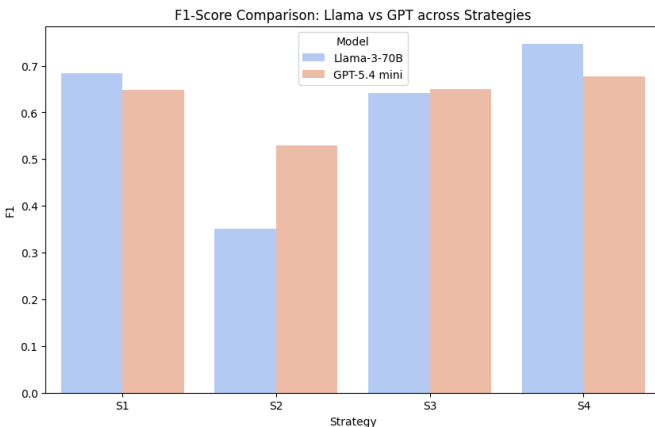


Fig. 1. Comparison of mean F1-scores of FB15k-237 and WN18RR across strategies S_1 – S_4 for Llama-3-70B and GPT-5.4 mini.

To further analyze the interaction between dataset complexity and prompting strategy, the F1-score distribution is visualized as a heatmap in Figure 2. This allows for immediate identification of performance peaks, such as the synergy between Llama-3-70B and the Expert Persona (S_4).

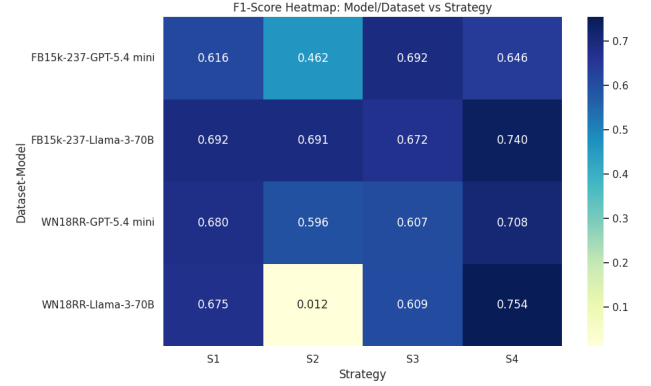


Fig. 2. Heatmap of F1-scores by Model, Strategy, and Dataset. Darker cells indicate higher factual validation accuracy.

Key Finding: Llama-3-70B using the Expert Persona (S_4) emerged as the overall winner in terms of F1-score on both datasets. While GPT-5.4 mini was more competitive in Zero-Shot scenarios, it did not scale as effectively as Llama-3-70B when provided with an expert role-play prompt.

B. Robustness Against Corruption Modes

To understand the “blind spots” of each model, we analyzed accuracy across specific corruption types using the top-performing strategy (S_4). The multi-dimensional performance across these modes—Head, Tail, and Relation corruptions—is represented in the radar charts in Figure 3.

- **Original Facts (Recall):** Llama-3-70B showed a superior recall for true facts, reaching 93.9% on WN18RR, whereas GPT-5.4 mini was more skeptical, correctly identifying only 71.3% of true WordNet relations.
- **Relation Corruptions:** This was the most difficult category for both models. Llama-3-70B struggled significantly with WordNet relation swaps (11.2% accuracy), while GPT-5.4 mini was more robust at 49.1%. This suggests GPT-5.4 mini has a deeper understanding of lexical ontological constraints.
- **Entity Corruptions (Head/Tail):** Both models performed well on tail corruptions for Freebase, but Llama-3-70B was notably better at identifying head corruptions in WordNet (85.2%) compared to its performance in Freebase (45.4%).

A comprehensive breakdown of corruption detection accuracies and recall across all four prompting strategies (S_1 – S_4) is provided in **Appendix E**. These tables highlight the significant variance in model behavior when transitioning from simple zero-shot queries to complex expert personas.

C. Prompt Sensitivity Analysis

The classification behavior and error types are detailed in the confusion matrices shown in Figure 4 and Figure 5. These matrices reveal the specific nature of model failures, such as the instability of output results stemming from the diversity of natural language expressions and the identification

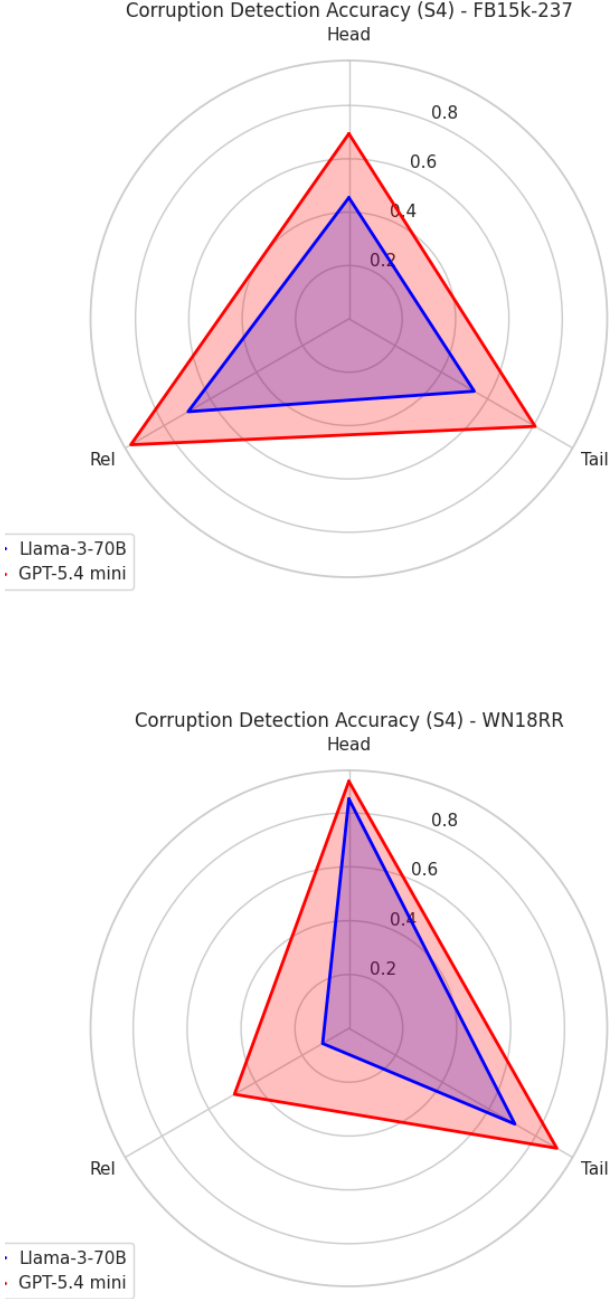


Fig. 3. Radar charts illustrating model accuracy across different corruption modes for FB15k-237 (top) and WN18RR (bottom). Performance on original facts represents the model’s Recall.

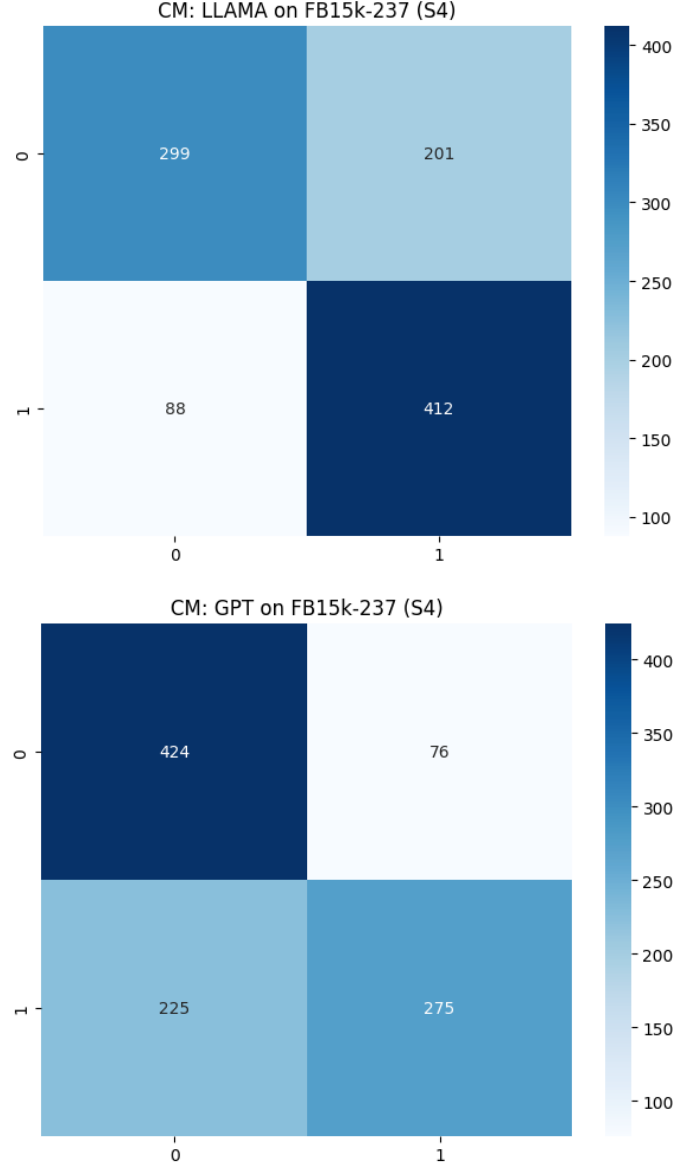


Fig. 4. Confusion matrices for FB15k-237 comparing Llama-3-70B (top) and GPT-5.4 mini (bottom). Note the distribution of errors between head and tail corruptions in the Freebase domain.

of incorrectly asserted relations that often reside within the ground-truth Knowledge Graph itself.

- **The Zero-Shot Affirmation Bias:** As hypothesized, S_1 (Zero-Shot) led to high Recall but low Precision for Llama-3-70B, indicating it often defaulted to “True.” GPT-5.4 mini exhibited significantly less bias in the Zero-Shot setting.
- **Few-Shot Divergence (S_2):** Few-shot prompting was catastrophic for Llama-3-70B on the WN18RR dataset (F1: 0.01), as the model began rejecting almost every triplet. In contrast, GPT-5.4 mini utilized few-shot examples effectively to stabilize predictions.
- **Expert Persona Synergy (S_4):** Assigning the “Knowledge Graph Engineer” persona was the most effective intervention for Llama-3-70B, raising its F1-score from 0.67 (baseline) to 0.75.

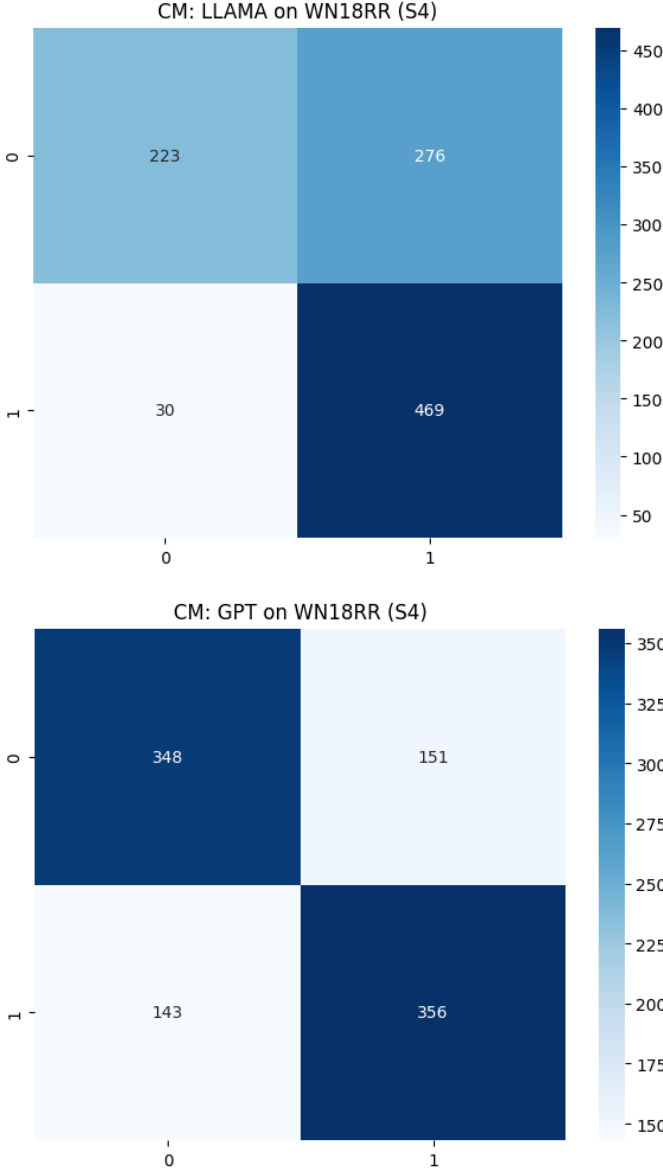


Fig. 5. Confusion matrices for WN18RR comparing Llama-3-70B (top) and GPT-5.4 mini (bottom). These matrices highlight the divergent handling of hierarchical lexical relations in the WordNet domain.

V. FAILURE TAXONOMY

Based on a deep analysis of the residual errors from the S_4 (Expert Persona) strategy, we identify four primary failure modes that constitute a failure taxonomy for LLM-based Knowledge Graph validation. This analysis reveals that failures are not stochastic but are rooted in specific cognitive and structural biases inherent to Large Language Models.

A. Category I: Ontological Granularity Over-Generalization

This failure mode is most prevalent in the WN18RR dataset, particularly under relation corruption. Models fail to distinguish between highly specific, semantically adjacent relations within a formal hierarchy.

- **Mechanism:** Models frequently collapse distinct lexical properties (e.g., `_member_meronym` vs. `_has_part`) into a generalized “semantic association” bucket.

- **Evidence:** Llama-3-70B achieved a striking accuracy of only 11.2% on WordNet relation corruptions. It consistently validated triplets like *enlist.v.03 [verb group] enlistment.n.01* as true, despite the relation being synthetically swapped.
- **Impact:** This suggests that while LLMs understand general association, they struggle with the strict logical constraints required for formal ontology maintenance.

B. Category II: The “Type-Safe” Plausibility Trap

Failures in this category occur when a model accepts a false triplet because the head, relation, and tail are semantically compatible, even if the fact is historically or factually incorrect.

- **Mechanism:** Because our corruption engine uses Type-Safe Candidate Pools, the negative samples are plausible. For example, replacing one actor with another in a film relationship often bypasses the model’s factual threshold if both actors are active in the same era.
- **Evidence:** In FB15k-237, Llama-3-70B’s accuracy on head corruption dropped to 45.4%. It incorrectly validated *South_Dakota [religion] Presbyterianism*, likely due to the high global popularity and plausible demographic association of these entities in the model’s pre-training data.
- **Significance:** This identifies a “Popularity Bias” where models prioritize statistical likelihood over precise factual verification.

C. Category III: Structural Skepticism and “Cynical” Rejection

In contrast to Llama’s optimism, GPT-5.4 mini exhibited a distinct failure mode characterized by extreme skepticism, particularly in the FB15k-237 domain.

- **Mechanism:** The model rejects ground-truth facts as false, resulting in high Precision but low Recall. It appears to penalize triplets that are complex, long-tail, or expressed through ambiguous natural language mappings.
- **Evidence:** GPT-5.4 mini correctly identified only 55.0% of true Freebase facts. It rejected valid triplets such as *Rambo: First Blood Part II [released in] DVD*, possibly due to a strict internal threshold for what constitutes a “verified” release relationship.
- **Contrast:** This represents the inverse of the “Affirmation Bias”; the model becomes an over-zealous filter, discarding valid knowledge and increasing the KG’s incompleteness.

D. Category IV: Semantic Ambiguity and Identifier Confusion

Both models struggle when natural language labels do not perfectly capture the underlying machine-readable logic of the KG.

- **Mechanism:** When an entity name is ambiguous (e.g., “Apple” the fruit vs. “Apple” the company) or when relations are represented by concatenated strings (e.g.,

award award ceremony awards presented), the models lose track of the specific directional logic.

- **Evidence:** We observed multiple failures in the WN18RR dataset involving `_derivationally_related_form`, where the models struggled to determine if the linguistic transformation was bi-directional or hierarchical, leading to a cluster of False Positives.

VI. TOPOLOGICAL ANALYSIS

To understand the structural consequences of LLM-based quality control, we measured the topological integrity of the “cleaned” sub-graphs—those consisting only of triplets validated as *True* by the models under the S_4 (Expert Persona) strategy. Table III summarizes the shifts in network architecture.

TABLE III

IMPACT OF LLM-DRIVEN GRAPH CLEANING ON TOPOLOGICAL METRICS

Version	Nodes	Edges	Density	Comp.	Avg Deg.
FB15k Baseline	1,055	750	0.00067	316	1.422
Llama-S4 Cleaned	840	546	0.00077	300	1.300
GPT-S4 Cleaned	578	341	0.00102	238	1.180
WN18RR Baseline	1,153	749	0.00056	404	1.299
Llama-S4 Cleaned	960	532	0.00058	428	1.108
GPT-S4 Cleaned	769	412	0.00070	357	1.072

A. The Density Paradox: Structural Refinement vs. Sparsification

A critical observation in Table III is the consistent increase in **Graph Density** despite a significant reduction in the total number of nodes and edges. In the FB15k-237 domain, GPT-5.4 mini reduced the edge count by 54.5%, yet the density improved from 0.00067 to 0.00102.

This suggests that the LLMs are not merely deleting random data; they are performing a “structural pruning” of noisy, weakly connected entities. The nodes being removed are often pendant nodes or isolated triplets that lack supporting context in the model’s pre-trained world knowledge. By removing these peripheral or misaligned statements, the models leave behind a more tightly coupled, high-veracity core. This supports the global perspective proposed in recent frameworks like Graphusion, where fusion and refinement lead to a more coherent global KG [37].

B. Model Skepticism and Graph Fragmentation

The two models exhibit divergent impact on the graph’s connectivity:

- **Conservative Curation (Llama-3-70B):** Llama-3-70B maintains a higher percentage of the original graph structure. However, in the WN18RR domain, the number of connected components **increased** from 404 to 428. This implies that Llama-3-70B correctly identified and removed “weak semantic bridges”—triplets that incorrectly linked disparate lexical clusters—thereby increasing the

graph’s fragmentation but improving its categorical purity.

- **Aggressive Filtering (GPT-5.4 mini):** GPT-5.4 mini acts as a high-precision filter, resulting in a significantly smaller graph. In FB15k-237, it reduced connected components to 238, indicating that it effectively collapsed the graph into its most prominent, well-supported factual hubs.

C. Implications for Downstream Applications

The topological shifts observed here have profound implications for the two primary ways Knowledge Graphs are utilized in modern AI:

1) Impact on Knowledge Graph Embeddings (KGE):

Traditional KGE models like TransE and RotatE rely on structural patterns to learn latent representations [30], [31]. The removal of noisy triplets, as identified by the LLM, directly mitigates the risk of “embedding pollution,” where incorrect edges pull entity vectors into incorrect regions of the complex space. The increased density in the cleaned graphs suggests that subsequent link prediction tasks would likely achieve higher hits@k scores, as the underlying training data is more semantically consistent [39].

2) Impact on Retrieval-Augmented Generation (RAG):

For RAG systems, the primary bottleneck is the presence of conflicting or hallucinatory facts within the retrieval source. By using the LLM as a curator to prune the KG, the probability of retrieving a “factually corrupt” triplet—such as a misaligned film director or a false lexical hypernym—is significantly reduced. This topological cleaning acts as a pre-emptive filter, ensuring that the input provided to a generative model is grounded in verified structure rather than extracted noise [34], [35].

VII. CONCLUSION

In this work, we demonstrated a scalable framework for KG quality refinement. Our findings suggest that open-source models like Llama-3, when properly persona-primed, can exceed the performance of proprietary frontier models for structured factual validation tasks.

VIII. FUTURE WORK

The findings of this study establish a performance baseline for general-purpose LLMs in Knowledge Graph (KG) validation. However, to achieve the level of veracity required for production-grade clinical or scientific systems, several emerging paradigms must be integrated.

A. Graph-Centric Supervised Fine-Tuning (SFT)

While zero-shot prompting strategies (S_1 – S_4) demonstrate the potential of LLMs, they often struggle with domain-specific noise and hallucination. Future work should transition toward the *Graph Judge* framework [42], where a model is explicitly fine-tuned on a corpus of verified and synthetically corrupted triplets. Such an approach enables the model to bypass the “plausibility trap” by learning the structural and semantic nuances of the target ontology through Knowledge-Aware SFT [42].

B. Agentic Multi-Model Consensus and Uncertainty

To address the affirmation bias observed in our results, we propose the implementation of a multi-agent retrieval-augmented generation (KG-RAG) pipeline [40]. This involves a consensus-based validation system where multiple models (e.g., Llama, GPT, and DeepSeek [48]) evaluate the same triplets using multi-agent prompting [49]. By incorporating entropy-based uncertainty scoring, the system can autonomously flag low-confidence or semantically inconsistent triplets for human-in-the-loop oversight, a strategy that has proven critical in high-stakes oncology domains [40].

C. From Local Extraction to Global Fusion

Current methodologies frequently suffer from a “local perspective” bottleneck, focusing on sentence-level triplets while ignoring global graph consistency [43]. Future iterations of this work should integrate *Graphfusion*, a framework that employs zero-shot fusion modules to merge local sub-graphs, resolve conflicting relationships across documents, and unify disparate entity mentions (e.g., deduplicating acronyms and their full forms) into a coherent global KG [43].

D. Dynamic Reasoning via Knowledge Graph of Thoughts (KGoT)

To improve the efficiency and cost-effectiveness of KG construction, we envision a shift toward the *Knowledge Graph of Thoughts* (KGoT) architecture [41]. By externalizing the model’s intermediate reasoning into a dynamic graph representation that is iteratively enhanced via external tools (e.g., math solvers and Python scripts), we can significantly improve success rates on complex benchmarks while reducing the hallucinations typical of pure autoregressive generation [41].

E. Interactive Refinement and Semantic Filtering

Finally, we aim to explore interactive multi-turn paradigms for Knowledge Base Question Answering (KBQA) [45]. By enabling models to interact directly with the KG through generic APIs and SPARQL-based feedback loops, the system can perform self-correction during the construction phase. This can be further bolstered by a two-stage filtering mechanism that balances external retrieval with the model’s intrinsic reasoning, mitigating the risks of excessive reliance on noisy retrieved data [44], [47]. Exploring these topological and network-based insights will facilitate a deeper understanding of complex disorders and scientific domains [46].

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APPENDIX

A. FB15k-237 Sampling Distribution

This subsection provides the relation distribution for the FB15k-237 dataset. Sampling was performed on 20,466 original triplets to produce a balanced evaluation set of 500 positive triplets. Table IV demonstrates the proportional representation of some relation types.

B. WN18RR Sampling Distribution

The sampling for WN18RR was performed on 3,134 original triplets. Table V demonstrates the proportional representation of all relation types ($N = 500$).

TABLE IV
FB15K-237 RELATION DISTRIBUTION (ORIGINAL VS. SAMPLED)

Relation ID	Original (N=20,466)		Sampled (N=500)	
	Count	Percentage (%)	Count	Percentage (%)
1	1,447	7.07	30	6.0
3	1,311	6.41	30	6.0
2	1,067	5.21	22	4.4
4	858	4.19	17	3.4
5	836	4.08	18	3.6
6	722	3.53	18	3.6
8	664	3.24	13	2.6
10	606	2.96	15	3.0
13	494	2.41	9	1.8
14	436	2.13	9	1.8

TABLE V
WN18RR RELATION DISTRIBUTION (ORIGINAL VS. SAMPLED)

Relation ID	Original (N=3,134)		Sampled (N=500)	
	Count	Percentage (%)	Count	Percentage (%)
_hypernym	1,251	39.92	200	40.08
_derivationally...	1,074	34.27	171	34.27
_member_mero..	253	8.07	40	8.02
_has_part	172	5.49	27	5.41
_instance_hype...	122	3.89	19	3.81
_synset_domain...	114	3.64	18	3.61
_also_see	56	1.79	9	1.80
_verb_group	39	1.24	6	1.20
_member_of_do...	26	0.83	4	0.80
_member_of_do...	24	0.77	4	0.80

C. Initial Network Topological Summary

Before synthetic corruption, the sampled sub-graphs were analyzed to characterize their structural baseline. Table VI summarizes these topological metrics.

TABLE VI
NETWORK TOPOLOGICAL METRICS FOR INITIAL SAMPLED TRIPLETS

Metric	FB15k-237	WN18RR
Total Entities (Nodes)	846	954
Total Relations (Edges)	500	499
Average Degree	1.18	1.05
Graph Density	0.00070	0.00055
Connected Components	350	455
Avg. Clustering Coefficient	0.0	0.0

D. Prompting Templates

The following standardized templates were used to query Llama-3-70B and GPT-5.4 mini. The variable {triplet} refers to the natural language mapping of the subject, predicate, and object.

Strategy S_1 (Zero-Shot Baseline)

Is the following factual triplet true? Answer only with 'Yes' or 'No'.

Triplet: {triplet}

Output:

Strategy S_2 (Few-Shot Learning)

Evaluate the factual accuracy of the final triplet based on the patterns shown in the examples.

Answer only with ‘Yes’ if true or ‘No’ if false.

Example 1: Paris [location contains] Eiffel Tower - Yes

Example 2: Albert Einstein [born in] Tokyo - No

Example 3: canine.n.01 [hypernym] dog.n.01 - Yes

Example 4: apple.n.01 [has part] engine.n.01 - No

Triplet: {triplet} -

Output:

Strategy S_3 (Chain-of-Thought)

Examine the following triplet and explain your reasoning step-by-step before concluding if it is true or false. Answer with ‘True’ or ‘False’ at the end.

Triplet: {triplet}

Output:

Strategy S_4 (Expert Persona)

You are a senior Knowledge Graph Engineer specializing in Freebase (real-world facts) and WordNet (lexical semantics). Your task is to validate the following triple for factual integrity.

Instructions:

1. Analyze the relationship between the head and tail entities.

2. Determine if this specific connection is recognized in a standard knowledge base.

3. Output exactly one word: ‘True’ or ‘False’.

Triplet: {triplet}

Output:

E. Detailed Corruption Detection Statistics

This section provides a granular analysis of detection accuracy across all corruption modes—Head, Tail, and Relation—alongside the recall (Original Facts) for every evaluated strategy.

FB15k-237 Detailed Performance

Table VII details the performance of both models on the Freebase dataset. Note the sharp drop in recall for GPT-5.4 mini in S_2 , indicating a extreme “skepticism” bias when few-shot examples were introduced.

TABLE VII

FB15K-237: ACCURACY BY CORRUPTION MODE AND STRATEGY

Model	Strategy	Head Acc.	Tail Acc.	Rel. Acc.	Recall
Llama-3 (70B)	S_1 (Zero-Shot)	14.9%	19.4%	18.8%	96.2%
	S_2 (Few-Shot)	50.4%	56.6%	77.6%	71.0%
	S_3 (CoT)	43.8%	38.8%	43.6%	79.8%
	S_4 (Persona)	45.5%	54.3%	69.6%	82.4%
GPT-5.4 (mini)	S_1 (Zero-Shot)	80.2%	79.8%	92.4%	50.6%
	S_2 (Few-Shot)	89.3%	90.7%	98.4%	31.8%
	S_3 (CoT)	50.4%	60.5%	71.2%	72.2%
	S_4 (Persona)	69.4%	80.6%	94.4%	55.0%

WN18RR Detailed Performance

Table VIII summarizes the results for the WordNet dataset. The catastrophic failure of Llama-3-70B in S_2 is evident here, where it rejected nearly all facts (Recall: 0.6%), resulting in artificially high detection accuracy for corruptions.

TABLE VIII

WN18RR: ACCURACY BY CORRUPTION MODE AND STRATEGY

Model	Strategy	Head Acc.	Tail Acc.	Rel. Acc.	Recall
Llama-3 (70B)	S_1 (Zero-Shot)	13.1%	17.2%	2.8%	97.2%
	S_2 (Few-Shot)	100.0%	100.0%	99.2%	0.6%
	S_3 (CoT)	59.0%	64.1%	41.0%	65.1%
	S_4 (Persona)	85.2%	71.1%	11.2%	94.0%
GPT-5.4 (mini)	S_1 (Zero-Shot)	95.1%	94.5%	54.6%	64.5%
	S_2 (Few-Shot)	98.4%	98.4%	84.3%	46.1%
	S_3 (CoT)	80.3%	85.2%	65.1%	54.9%
	S_4 (Persona)	91.8%	89.1%	49.0%	71.3%